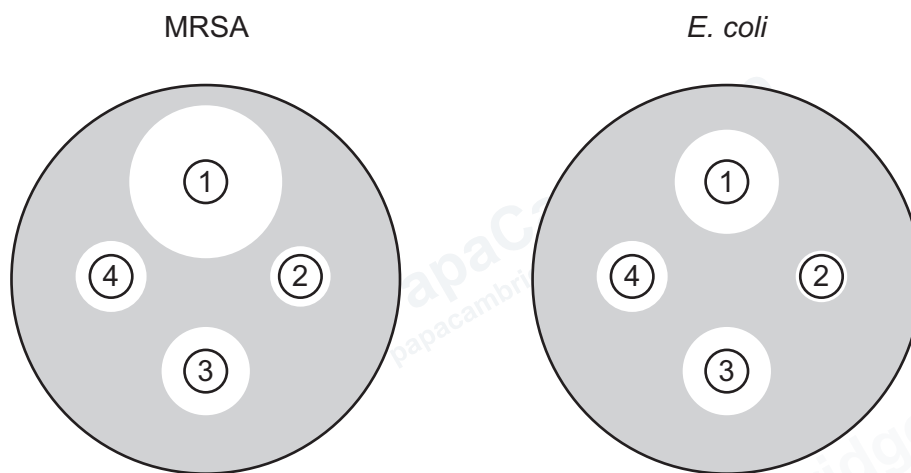


- 37 Scientists were tasked with finding alternative treatments to antibiotics for MRSA (Methicillin resistant *Staphylococcus aureus*) and *Escherichia coli* (*E. coli*).

MRSA and *E. coli* were grown on separate agar plates with four filter paper discs that had been soaked in one of the $200\ \mu\text{g cm}^{-3}$ solutions listed.

- 1 garlic extract
- 2 silver nanoparticles
- 3 gentamicin antibiotic
- 4 curcumin extract

Diagrams of the agar plates after a 48-hour incubation period are shown. The shaded areas on the agar plates show the location of living bacteria.



Which statement is a correct conclusion of the results?

- A Gentamicin antibiotic can be used in combination with garlic extract to treat MRSA and *E. coli* infections.
- B Garlic extract can be used to replace antibiotics in the treatment of MRSA and *E. coli* infections.
- C MRSA and *E. coli* are affected more by garlic extract than treatments 2, 3 and 4.
- D Treatments 2 and 4 have **no** effect on MRSA and *E. coli* infections.